CSE 421 ID: 24241206

Application Layer Protocols (HTTP.SMTP/POP) Examination Lab

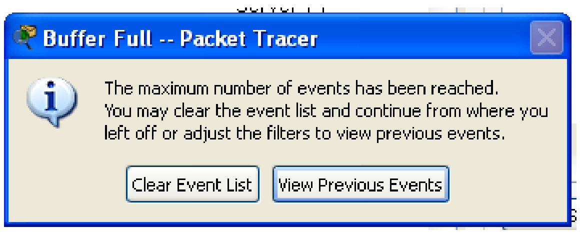
Objectives:

Capture traffic and observe the PDUS for HTTP, SMTP, POP.

Task 1: Observe HTTP traffic exchange between a client and server.

Step 1 – Run the simulation and capture the traffic.

- Enter **Simulation** mode.
- Click on the PC1. Open the **Web Browser** from the **Desktop**.
- Enter **www.bracu.ac.bd** into the browser. Clicking on **Go** will initiate a web server request. Minimize the Web Client configuration window.
- Two packets appear in the **Event List**, a DNS request needed to resolve the URL to the IP address of the web server and an ARP request needed to resolve the IP address of the server to its hardware MAC address.
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.



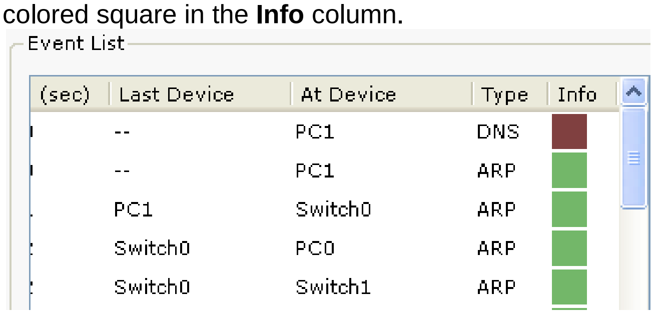
- When the above message appears Click "View Previous Events".
- Click on PC1. The web browser displays a web page appears.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe HTTP traffic.

	Last Device	At Device	Type
1.	PC1	Switch 0	HTTP
2..	Local Web Server	Switch 1	HTTP

- Find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.



- When you click on the Info square for a packet in the event list the **PDU Information** window opens. If you click on these layers, the algorithm used by the device (in this case, the PC) is displayed. View what is going on at each layer.

- Examine the PDU information for the remaining events in the exchange.

For packet 1::

What kind of HTTP packet is packet no. 1?

HTTP Request packet

Click onto "Inbound PDU details" tab. Scroll down at the end, what do you see?

HTTP request header information is shown.

These headers indicate the language preference, the types of data the client can accept, the connection status, and the hostname of the requested web server.

HTTP Data:Accept-Language: en-us
Accept: */*
Connection: close
Host: www.bracu.ac.bd

For packet 2:

Click onto "Inbound PDU details" tab. Scroll down at the end, what do you see? What kind of HTTP packet is this?

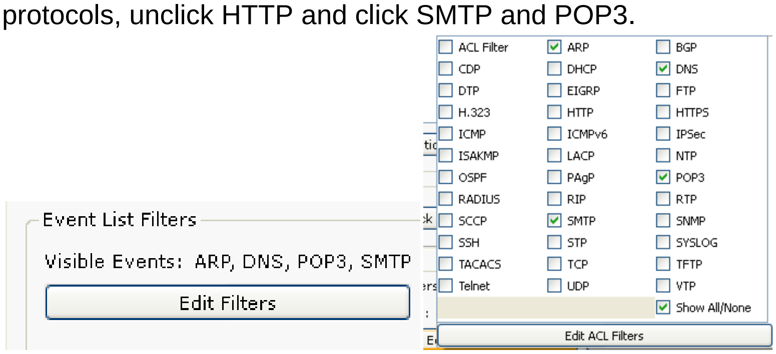
HTTP Response packet

HTTP Data:Connection: close
Content-Length: 151
Server: PT-Service/5.2

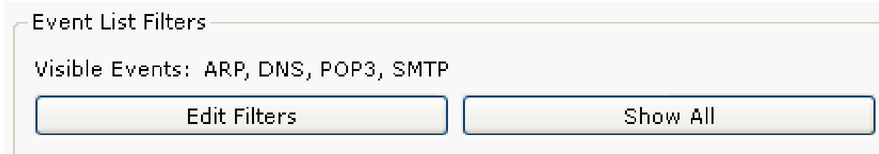
Task 2: Observe email traffic exchange between a client and email server using SMTP and POP3.

Step 1 – Run the simulation and capture the traffic.

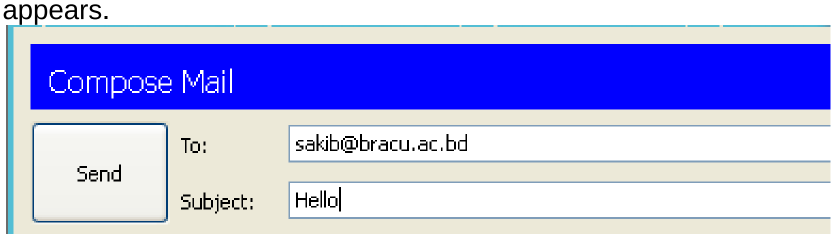
- On the Event List window click "Reset Simulation" button. All previous packets will disappear.
- At the bottom of the Event List window, there is a filter which filters the protocols that we want to see. Click Edit filters. Another window appears showing different protocols, unclick HTTP and click SMTP and POP3.



- Click a space anywhere outside the popup window, then it will disappear.
- Your Event List Filter should be as shown below:



- Now click on the PC1. Close the web browser window. Open the **Email** from the **Desktop**. A mail browser window will open. Click “compose”, another window appears.



- Fill the window as shown and press send.
- Minimize the client window .
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.
- This interaction is between the sender client and its email server.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe SMTP traffic.

	Last Device	At Device	Type
3.	PC1	Switch 0	DNS
4.	PC1	Switch 0	SMTP
5.	Bracu Email Server	Switch 1	SMTP

- Find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.
- Examine the PDU information.

For packet 4::

What is the purpose of this DNS packet?

To get the IP address from the domain names.

The client needs the IP address so that it can establish communication with the email server and send the email.

For packet 5& 6::

Explain why SMTP packet was sent to the email server and the server replied with an SMTP packet?

The SMTP packet was sent from the client to the email server to transfer the composed email message to the mail server. SMTP is responsible for sending outgoing email. SMTP is a push protocol

The server replied with an SMTP packet to acknowledge the receipt of the email and confirm that the message has been accepted for delivery.

Step 3 – Run the simulation and capture the traffic for POP.

- On the Event List window click "Reset Simulation" button. All previous packets will disappear.
- Now click on the PC0. Open the **Email** from the **Desktop**. A mail browser window will open. Click "**receive**", minimize the window.
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.
- This interaction is between the sender client and its email server.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe POP traffic.

	Last Device	At Device	Type
6.	PC1	Switch 0	DNS
7.	PC1	Switch 0	POP3
8.	Bracu Email Server	Switch 1	POP3

- Find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.
- Examine the PDU information.

For packet 6::

What is the purpose of this DNS packet?

To find the IP address of the email server from its domain name.

This allows the client to connect to the correct mail server in order to retrieve emails.

For packet 7&8::

Explain why POP packet was sent to the email server and the server replied with a POP packet?

The POP3 packet was sent from the client to the email server to request retrieval of stored emails from the mailbox. POP3 is a pull protocol.

The email server replied with a POP3 packet to send the requested email messages and confirm the successful retrieval of emails by the client.

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